03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 10

(JCD 6.3 - 7)

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Previous lecture

- Convection happens when absolute temperature gradient is high
- Ledoux criterion:

$$\frac{d \ln \rho}{d \ln P} < \frac{1}{\Gamma_1}$$

- Schwarzschild criterion:

$$\nabla = \frac{d \ln T}{d \ln P} > \frac{\Gamma_2 - 1}{\Gamma_2} = \nabla_{\mathrm{ad}}$$

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Outline

Occurrence of convection

Modeling convection

Mass-Luminosity relation

Summary

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The radiative temperature gradient

- In the Schwarzschild criterion, we used ∇, which we can call the radiative temperature gradient when energy is transported by radiative transfer.

$$\nabla = \nabla_R = \frac{d \ln T}{d \ln P} = \frac{dT}{dr} \frac{dr}{dP} \frac{P}{T}$$

- Use the equation of radiative transport, the equation of hydrostatic equilibrium, and the ideal gas law to show that

$$\nabla_R = \frac{3k}{16\pi \, a \, c \, G \, m_u} \frac{\kappa}{\mu} \frac{L(r)}{M(r)} \frac{\rho}{T^3}$$

- The instability condition is thus

$$\nabla_R > \nabla_{\text{ad}}$$

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When does convection occur

$$\boxed{\nabla_R = \frac{3k}{16\pi\, a\, c\, G\, m_u} \frac{\kappa}{\mu} \frac{L(r)}{M(r)} \frac{\rho}{T^3} > \nabla_{\text{ad}}}$$

- Large opacity κ - from Kramers approximation, this happens for lower temperatures, eg atmospheres of cool stars.
- Large luminosity to mass ratio $L(r)/M(r)$ - eg in massive star interiors (see later when we cover energy generation).
- Large density to temperature ratio ρ/T^3 - eg in photospheres of stars, typically lower-mass (cooler) stars.
- Small adiabatic temperature gradient ∇_{ad} - eg cool star atmospheres

Convection zones for main sequence stars

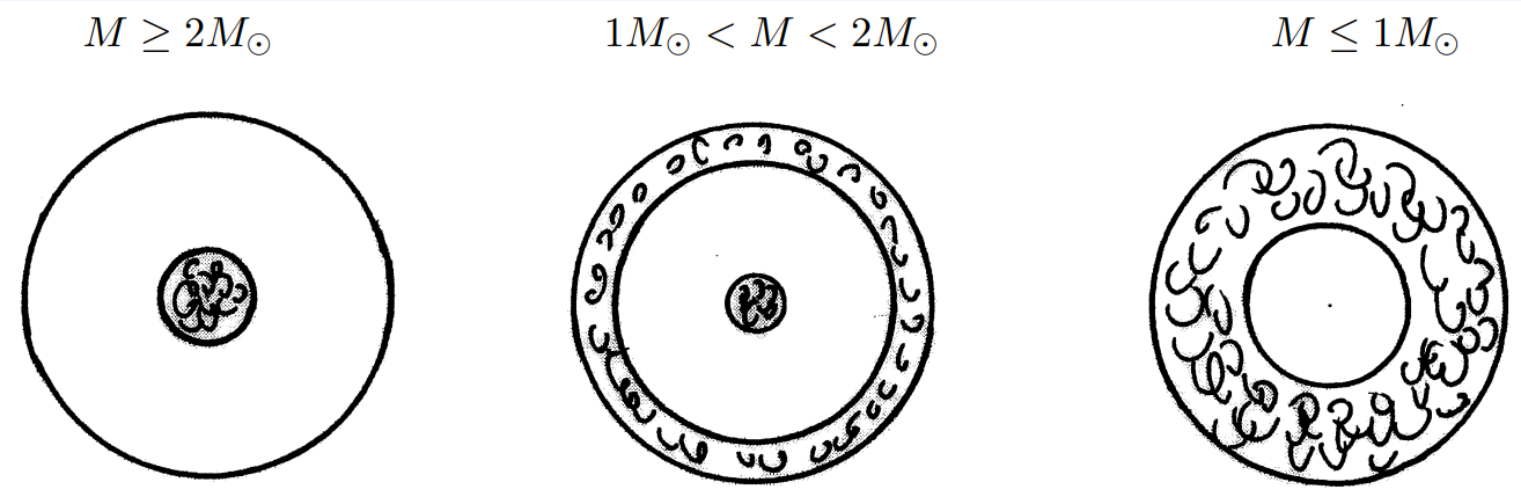
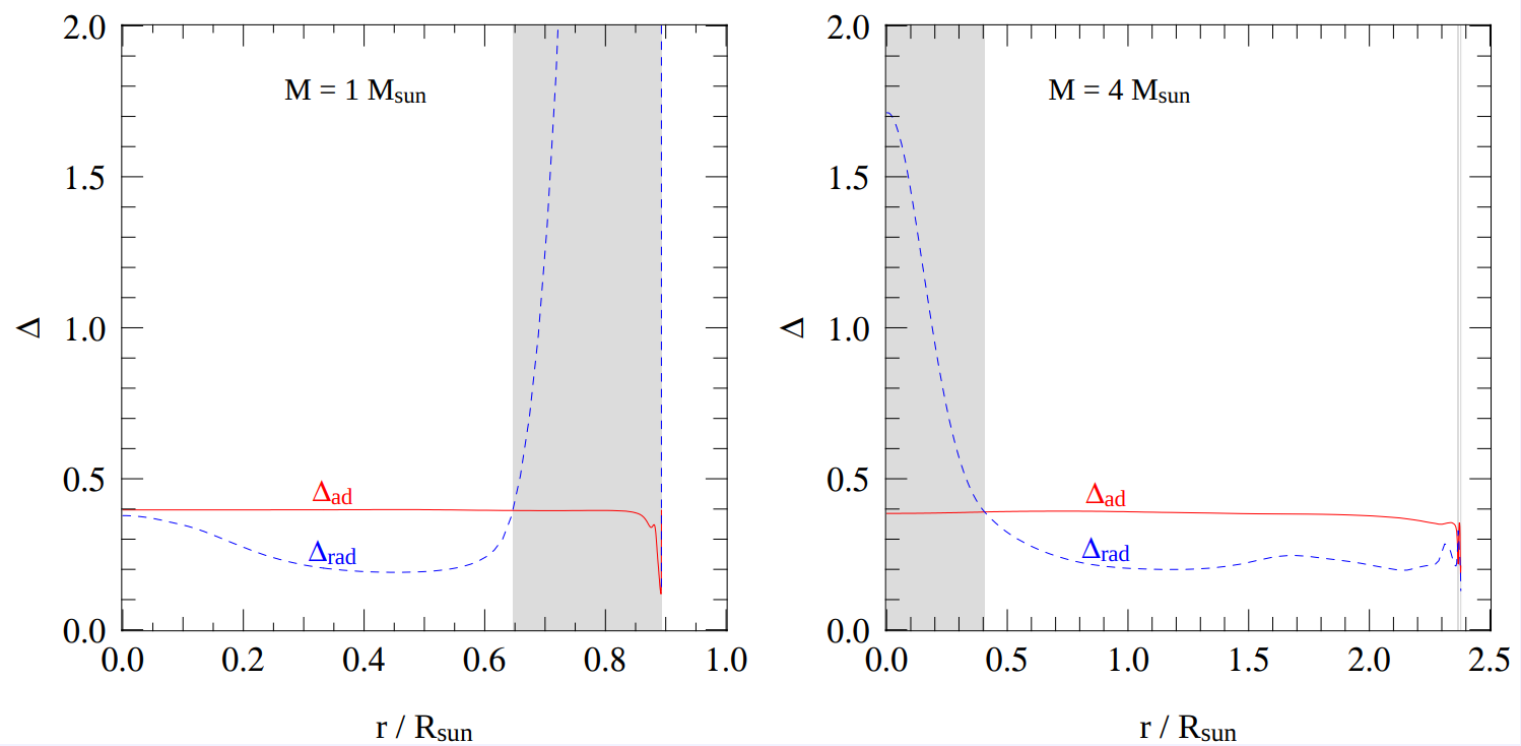


Image credit: JCD

- There are radiative and convective zones. The structure is mainly dependent on stellar mass.
- Note: Some very cool stars (cool M dwarfs) can be fully convective.
- As convection is less well understood than radiation, stellar models for cooler stars are often not accurate.

Convection zones for main sequence stars



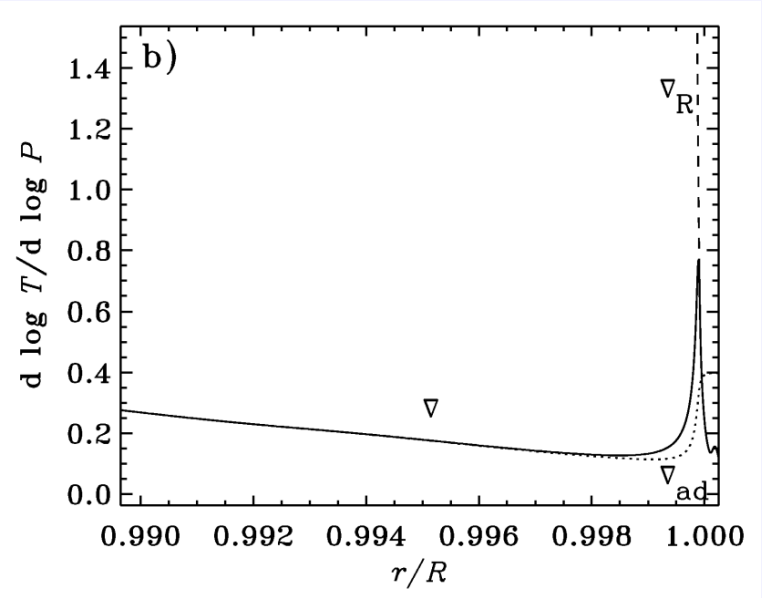


Image credit: Pols & JCD

Outline

Occurrence of convection

Modeling convection

Mass-Luminosity relation

Summary

Modeling convection

- Theory of convection still very active field of research.
- Full hydrodynamical simulations across length and time often unfeasible.
- Simplified theory is mixing length theory (MLT) - and it is not simple at all (see JCD6.4.2 for further simplification).
- MLT treats the complex convective motions as blobs of gas travelling over a distance l_m, the mixing length, and then dissolve.
- The mixing length can be expressed in terms of the pressure scale height.
- At the boundaries of convective and radiative zones, convective overshooting can happen.

Convective timescale

- The convective timescale (t_{con}) lies between the dynamical and thermal timescale.
- For the Sun, it is about $0.2\,\mathrm{yr} \approx 70\,\mathrm{d}$.
- Over a time scale not much longer than t_{con} matter in a convection zone must be completely mixed.
- We thus assume that convection zones are chemically homogeneous.

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Stars dominated by radiative transport

- We can estimate stellar luminosity - assumed roughly constant throughout a star - by using before-used equations and estimates.
- Use the temperature estimate from lecture 5, the equation of radiative transport from lecture 7.
- Estimate opacity as $\kappa \approx \kappa_0 \rho^{\lambda} T^{-\nu}$
- You can now show that

$$\boxed{L_* \approx \frac{a\,c}{\kappa_0} \left(\frac{G\, m_u\, \mu}{k} \right)^{4+\nu} R^{3\lambda-\nu} M^{3+\nu-\lambda}}$$

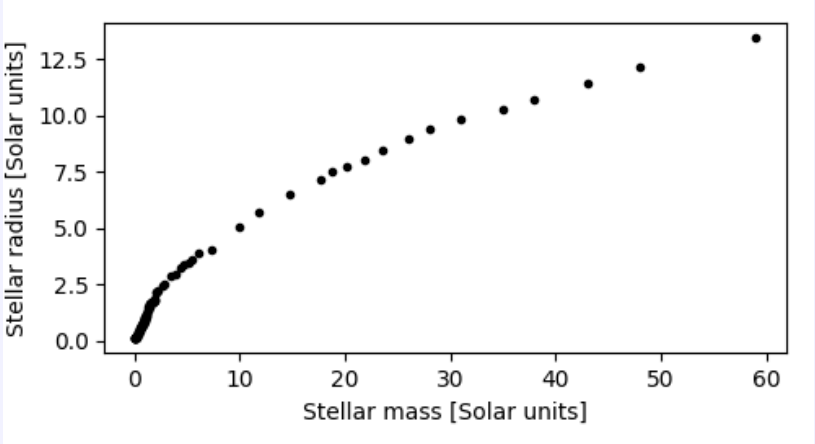
Lower main sequence stars

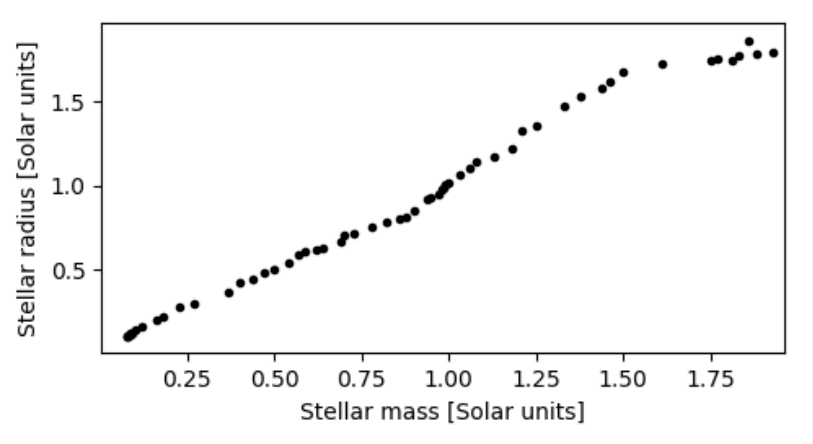
$$L_* \approx \frac{a\,c}{\kappa_0}\left(\frac{G\,m_u\,\mu}{k}\right)^{4+\nu} R^{3\lambda-\nu} M^{3+\nu-\lambda}$$

- For lower main sequence stars, opacity mainly happens through bound-free transtions (see lecture 8).
- Thus, we can use Kramers approximation and hence $\lambda = 1$ and $\nu = 3.5$
- Applying that to the general estimate gives

$$L_{*,\text{low mass MS}} \propto M_*^{5.5} R_*^{-0.5} \mu^{7.5}$$

Lower main sequence stars





(Using the Pecaut & Mamajek 2013 results)

- For lower main sequence stars, $M \propto R$
- Thus, mass dependence dominates:

$$\boxed{L_{*,\text{low mass MS}} \propto M_*^5 \mu^{7.5}}$$

Upper main sequence stars

$$L_* \approx \frac{a\,c}{\kappa_0}\left(\frac{G\,m_u\,\mu}{k}\right)^{4+\nu} R^{3\lambda-\nu} M^{3+\nu-\lambda}$$

- For upper main sequence stars, opacity mainly happens through electron scattering (see lecture 8).
- Thus, opacity is constant and hence $\lambda = 0$ and $\nu = 0$
- Applying that to the general estimate gives

$$\boxed{L_{*,\text{high mass MS}} \propto M_*^3 \mu^4}$$

- The mass dependence still dominates.

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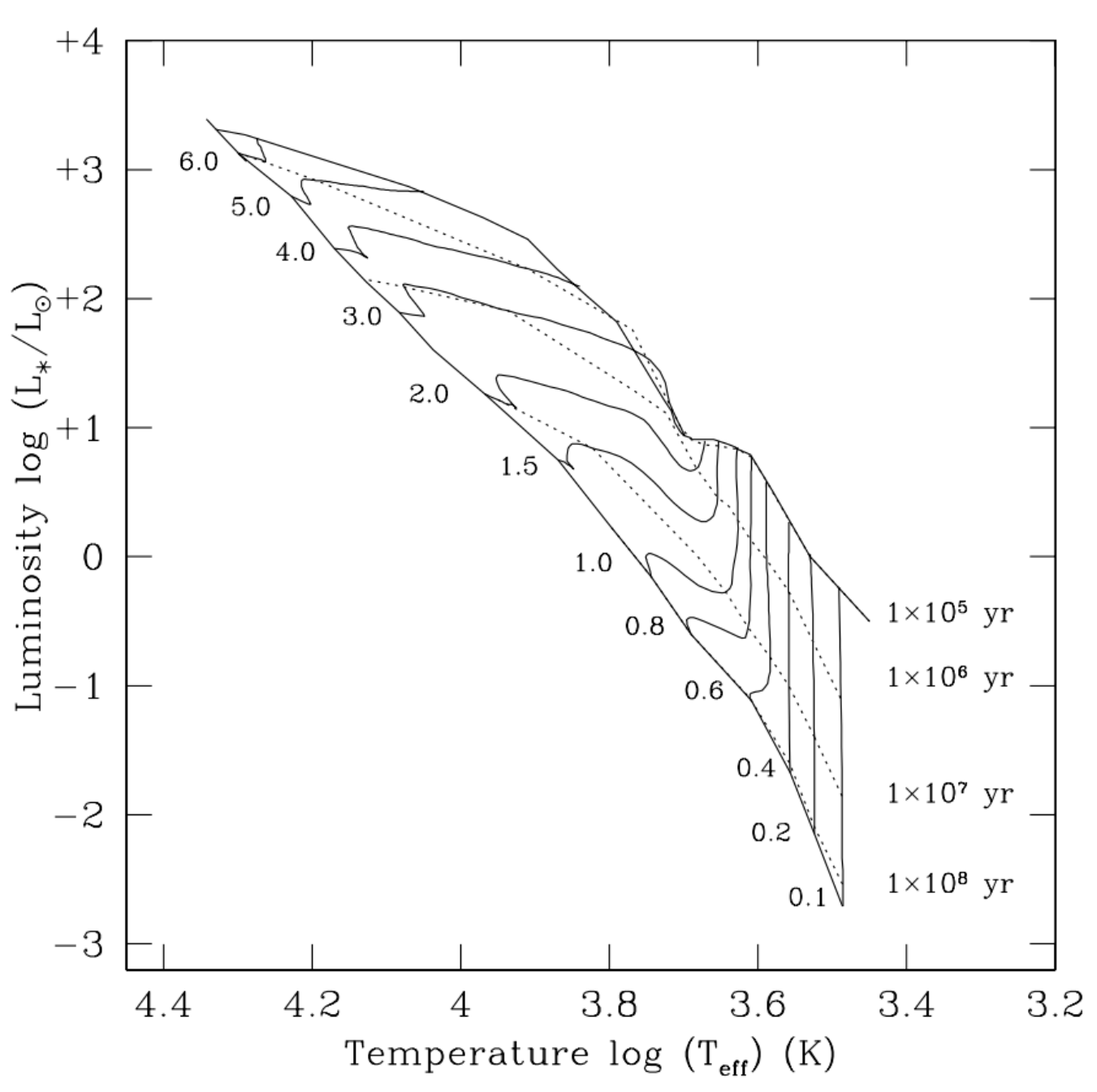
Stars dominated by convection

- Coolest stars are dominated by convective zones.
- Deriving a mass-luminosity relation is trickier. The main (and valid) assumption is that these stars behave as polytropes (see lecture 6).
- A reasonable relation is

$$L_{*,\mathrm{convective}} \propto M_*^{-7} T_{\mathrm{eff}}^{40}$$

- Note that most cool stars have similar effective temperatures ($\sim 2400\,\mathrm{K}$).
- From the relation between luminosity, radius and effective temperature, this then means that such stars decrease in luminosity with decreasing radius during their formation - such a vertical track in the HR diagram is called the Hayashi track.

Hayashi tracks and the forbidden region



(Image credit: JCD)

Summary

- Convective regions are mass-dependent:
  - Massive stars: convective core, radiative envelope.
  - Intermediate stars: convective envelope and core with radiative zone in between.
  - Low-mass stars: Convective envelope and radiative core.
  - Coolest, lowest-mass stars: Fully convective
- M-L relation also mass-dependent:
  - Lower Main sequence: $L_{*,\text{low mass MS}} \propto M_*^5 \mu^{7.5}$
  - Upper main sequence: $L_{*,\text{high mass MS}} \propto M_*^3 \mu^4$
  - Fully convective stars: $L_{*,\text{convective}} \propto M_*^{-7} T_{\text{eff}}^{40}$

NEXT LECTURE: Nuclear energy generation, cross sections, energy generation rate